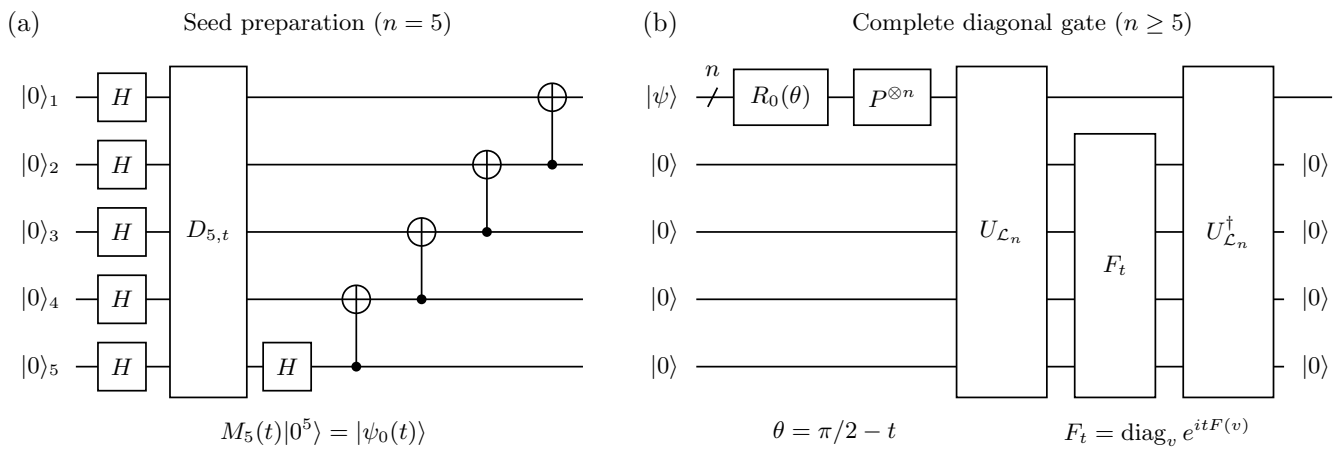

(a)
Seed preparation ($n = 5$)
$|0\rangle_1$
$|0\rangle_2$
$|0\rangle_3$
$|0\rangle_4$
$|0\rangle_5$
H
$D_{5,t}$
$M_5(t)|0^5\rangle = |\psi_0(t)\rangle$
(b)
Complete diagonal gate ($n \geq 5$)
$|\psi\rangle$
n
$R_0(\theta)$
$P^{\otimes n}$
$|0\rangle$
$U_{\mathcal{L}_n}$
F_t
$U^\dagger_{\mathcal{L}_n}$
$\theta = \pi/2 - t$
$F_t = \mathrm{diag}_v\, e^{itF(v)}$